

Lackey Clinic: Insight Brief

Team 6

Machine Learning 1 (Fall 2025)

William & Mary

Executive Summary

Bottom Line Up Front – Lackey Clinic demonstrates a strong ability to serve a community linked by unifying financial realities while embracing the diversity of their backgrounds and lifestyles. Economic constraints associated with a household income of three times the poverty level or less add layers of complexity to the analysis and modeling of factors that may contribute to patients leaving clinic care.

Sixth Sense Analytics finds two distinct “churn” groups at Lackey Clinic: patients who onboard but never engage, and patients who disengage after some time. After carefully analyzing and modeling the data, we find that people who churn do not have easily identifiable characteristics or patterns associated with their departure. Although we struggled to find factors to identify potential churn patients, we identified several “protective” factors, or things that the patients who stay often have in common.

When we started this analytic journey, we asked, “How do we solve the churn problem?” We believe the proper question is “What successes can we leverage to increase patient retention?” To some, it may seem like semantics, but we believe Lackey has found success and can grow that success by doing more of what they do best – meeting clients where they are and engaging with them as much as they will accept.

Problem Definition

Lackey Clinic is a world-class provider of healthcare services in Williamsburg, Yorktown, and the surrounding areas of Virginia. Their published mission statement is “To provide skilled, compassionate healthcare and counseling to the medically disadvantaged in a manner that honors the name of Jesus Christ.” As such, they seek to understand and minimize factors contributing to client disengagement from Lackey Clinic and to disruptions in the continuity of care. In many industries, churn refers to customers ceasing to do business with a company. Lackey Clinic uses this term to refer to the disengagement mentioned above.

Lackey wants a reliable way to identify patients at risk of churn, so that they can intervene before a patient drops out of care, hopefully avoiding a gap in

healthcare coverage. Churn, for analytical purposes, needs refinement. Our team initially defined churn as “a patient who has not re-enrolled with and has not had a visit to the clinic in the past 365 days.” When we started our initial analysis, we quickly saw that the number of people who churned was much higher than those who did not. This made it seem like Lackey could not keep patients and had a huge problem. We re-evaluated our definition after some data cleaning and more analysis. We also determined that there are two types of churn: 1) churn from Lackey Clinic, with a transition to something else that maintains continuity of care. 2) churn from Lackey into an unknown situation or no plan for healthcare coverage. This helped us arrive at a definition that allowed us to focus on the second type of churn, which we dubbed “bad churn”. Our definition of bad churn is as follows:

- The patient initially enrolled after January 01, 2021.
- The patient did not re-enroll at the end of an annual period.
- The patient had an “Unknown” or “Expired” disenrollment reason after data cleansing.

Methodology

After selecting our churn definition, we filtered the data to include only enrollments and churn events from 2021-01-01 to 2025-09-30. Our team chose this end date because it marked the end of our first data pull, and if we received updated data, we did not want to have an excess of null values when joining the tables.

The reason for that particular start date is more complicated. We noticed that older data seemed less standardized. We assumed, and later confirmed with the clinic, that before the COVID-19 pandemic, most enrollments were completed on paper and entered into their computer system manually. COVID forced people to go paperless, which required them to fill out an online form. This led to data standardization and mandatory entry of many fields. We made an educated guess that January 1, 2021, was far enough into the pandemic to be a good starting point.

Many reasons for disenrollment were already present in the data and revealed numerous explanations for transitioning to care elsewhere. The fact that they had care elsewhere was a good thing. Sometimes they moved out of the area or died. While not exactly good, care could not continue with

Lackey, so intervention was not possible, and these individuals were also excluded from the “bad churn.”

The data we used was a series of CSV files containing de-identified Lackey patient data provided by our professor. Specific data that we looked at and considered for analysis and modeling is:

- Patient personal life specific data – age, gender, marital status, housing situation, zip code, and length of time at that location, language, employment status, race, and education level.
- Patient clinic encounter-specific data: visited provider specialty, appointment completion status, appointment observations (e.g., BMI, PHQ-9, blood pressure readings), diagnoses, and orders.

The People, Enrollment, and Residence data required extensive cleaning to standardize and consolidate entries for analysis.

Our approach was to let the data speak for itself before we started asking questions. To that end, we conducted a comprehensive analysis of almost all the features in the data, including employment, marital status, education, number of appointments, and more. Similar to using different methods, we focused on areas with significantly higher or lower churn than the average. For example, individuals who speak Spanish churn at 14.4% which is less than

We ran multiple models, including GLM, LASSO, LDA, and Random Forest, to balance interpretability and performance. Of the models we ran, GLM and Random Forest performed best at predicting churn, though not as well as we had hoped.

Key Findings

Finding 1 - None of the models are really good at predicting who will churn

Our best-performing model achieved a 71.8% Sensitivity to churn, 48.8% Specificity, and 54% accuracy. In practical terms, this means it can catch 89 Churners, miss 35 Churners, and have 216 False Alarms. It is slightly better than guessing, but not reliable for individual patient prediction. For the onboarding segment, models performed at ~54%—essentially random chance.

This means that easily identifiable patient characteristics do not drive churn at Lackey. The population shares financial constraints and life complexities that make individual prediction difficult.

Finding 2 - Patients who make it past their third appointment churn less

The most evident pattern we saw was that engagement momentum prevents churn. Once patients complete three or four visits, they are more likely to stay.

Encounter Volume	Churn Rate	% of Churners	Lift	Priority
Low (1-3 visits)	32.4%	37.1%	1.43	CRITICAL
Medium (4-10)	25.1%	31.2%	1.11	Monitor
High (11-20)	18.7%	16.8%	0.83	Lower
Very High (>20)	13.0%	14.8%	0.58	Protective

New patients churn at 31.1% vs. 19.1% for recertifying patients. New patients represent 40% of all Churners. Getting them to their first appointments and establishing a connection is crucial for non-episodic patients.

Finding 3 - Spanish-speaking patients and people using mental health services churn less.

Spanish-speaking patients churn at nearly half the rate of English speakers (14.4% vs. 26.5%). Our GLM model confirms this is statistically significant ($p < 0.001$). Spanish speakers are under-represented among Churners by 12.5 percentage points.

Patients with elevated PHQ-9 scores (mental health screening) also show lower churn (15.9%). Those engaged with behavioral health services are staying in care.

This means that something in how Lackey serves these populations is working. The decreased churn rate may reflect language-concordant care that fosters stronger relationships, cultural factors that support engagement, or mental health services creating meaningful connections.

Finding 4 - Some Patterns Defy Simple Intervention

We identified groups with elevated churn for which the reasons may not be actionable. Some interesting data points from the chart below are:

- Age 20-29 (28.9% churn): Young adults may be episodic users—addressing an immediate need and moving on. This "one and done" pattern reflects life stage, not service failure.
- Age 65+ (32.0% churn): Patients approaching Medicare eligibility transition to other coverage. This is arguably success, not failure.
- Associate Degree holders (46.9% churn): This small group (32 patients) may be upwardly mobile individuals whose education leads to employment with benefits.

[Retention] ORM Risk Matrix: All Elevated Categories

Probability (Lift) × Magnitude (Impact) | Impact threshold: 0.186

Feature	Category	Lift	% Churners	Impact	Priority ¹
Patient Type	New Patient	1.37	40.0%	0.55	CRITICAL
Encounter Volume	Low (1-3)	1.43	37.1%	0.53	CRITICAL
Age Range	20-29	1.28	20.1%	0.26	CRITICAL
Age Range	65+	1.41	14.4%	0.20	CRITICAL
Language	English	1.17	77.5%	0.91	Volume Play
Telehealth Use	No Telehealth	1.27	31.4%	0.40	Volume Play
Housing	Lives with others	1.20	27.6%	0.33	Volume Play
No-Shows	1 No-Show	1.15	28.4%	0.33	Volume Play
Clinic Tenure	2-3yr	1.27	15.6%	0.20	Volume Play
Telehealth Use	Very High (>75%)	1.46	8.3%	0.12	Monitor
Race	Two or more races	1.58	5.3%	0.08	Monitor
Employment	Retired	1.30	5.1%	0.07	Monitor
Education	Associate Degree	2.07	3.0%	0.06	Monitor
County	Poquoson	1.31	2.6%	0.03	Monitor
Behavioral Health Dx	1	1.28	13.6%	0.17	Lower Priority
Telehealth Use	High (51-75%)	1.23	8.1%	0.10	Lower Priority
No-Shows	3-5 No-Shows	1.25	6.5%	0.08	Lower Priority
Race	Chooses not to identify	1.18	3.4%	0.04	Lower Priority

¹ CRITICAL = High probability + High impact (address immediately)

Recommendations

Recommendation 1 - Focus on the first three appointments

The data is unambiguous: patients with 1-3 visits churn at 32.4%; patients with 4+ visits churn at 13-25%. Concentrate retention resources on the critical first appointments.

Intervention	Timing
Schedule the next appointment before the patient leaves	Every visit
Confirmation call or text message before each visit, but especially those first three.	48 hours prior
"New patient pathway" check-ins	30, 60, 90 days
Flag patients who haven't returned after the first visit	45-day trigger
Consider "third visit milestone" acknowledgment	At the third visit

Recommendation 2 - Investigate Spanish-Speaking Patient Success

Spanish speakers churn at 14.4% vs. 26.5% for English speakers. Whatever is happening here should be understood and extended to all patients. There could be something cultural at play that is not replicable outside this patient population, but we will not know until more analysis is done.

- Interview the front-desk and clinical staff about service differences
- Review whether bilingual staff create stronger patient-provider relationships
- Assess if appointment reminders or follow-up processes differ
- Consider applying successful relational practices clinic-wide

Recommendation 3 - Strengthen Mental Health Engagement Pathways

Patients with elevated PHQ-9 scores and behavioral health engagement show lower churn. Mental health services appear to create meaningful connections that encourage ongoing engagement.

- Ensure PHQ-9 screening happens consistently at intake and annual visits
- Proactively connect patients with elevated scores to behavioral health
- Consider whether behavioral health's relational model could extend to primary care
- During our site visit, on-site "reverse psychology" visits were mentioned. Investigate if other areas of care might benefit from a similar "reverse" situation.

Recommendation 4 - Universal Onboarding Outreach

34% of enrolled patients never attend a visit, and no demographic factors predict who will fail to engage. The enrollment-to-first-visit process itself needs examination.

- Call all new enrollees within 48-72 hours of enrollment
- Assess barriers: transportation, scheduling conflicts, understanding of services
- Offer telehealth as an alternative when in-person barriers exist
- Track conversion rates and time-to-first-visit as key metrics
- Add a few enrollment questions that assess how comfortable a patient is with using their phone for app installation, text messaging receipt and response, and video calling.

Limitations

Our models achieved around 71% accuracy for retention and about 54% for onboarding. As stated in our findings, this is meaningful for identifying population patterns but not good enough for individual patient prediction. Sadly, there is no magic identifier that lets us see who exactly needs focused attention.

Data we did not have but might have been helpful includes the following:

Life events - job changes, family trauma events, housing instability changes. These would be useful in real time but there would need to be a way for patients to be encouraged to share that information on their own.

Care experience - patient satisfaction survey, wait times in the waiting room and pharmacy, appointment lead time, and other things that may be frustrating for patients and cause them to look elsewhere.

Transition Outcomes - for those that we labeled as bad Churners, finding out what happened to them. We suspect the number of bad Churners would decrease if we could engage those who left. People might not feel obligated to explain why they chose to leave Lackey Clinic when they leave. However, they may be willing to do so later, after some time has passed since the departure event.